

03 Rolling Validation Selection Error Rate (Expanding Window)

Aileen Tao

Linear truth

Simulation settings

Table 1: Simulation settings (expanding window, linear truth).

setting	value
seed	2026
Monte Carlo replications per n	500
Sample sizes (grid)	50, 100, ..., 1000 (step 50)
truth type	linear
beta0	1
beta1	2
beta2	NA
beta3	NA
noise type	gaussian
sigma	2
expanding initial window size	20
horizon	1
validation step	1

Error rate summary

Table 2: Monte Carlo mean selection-error rate (truth is linear; error = poly chosen) and approximate MC standard error.

n	error_rate	mc_se
50	0.006	0.00345
100	0.000	0.00000
150	0.000	0.00000
200	0.000	0.00000
250	0.000	0.00000
300	0.000	0.00000
350	0.000	0.00000
400	0.000	0.00000
450	0.000	0.00000
500	0.000	0.00000
550	0.000	0.00000
600	0.002	0.00200
650	0.000	0.00000
700	0.000	0.00000
750	0.000	0.00000
800	0.000	0.00000
850	0.000	0.00000
900	0.000	0.00000
950	0.000	0.00000
1000	0.000	0.00000

Error rate vs sample size

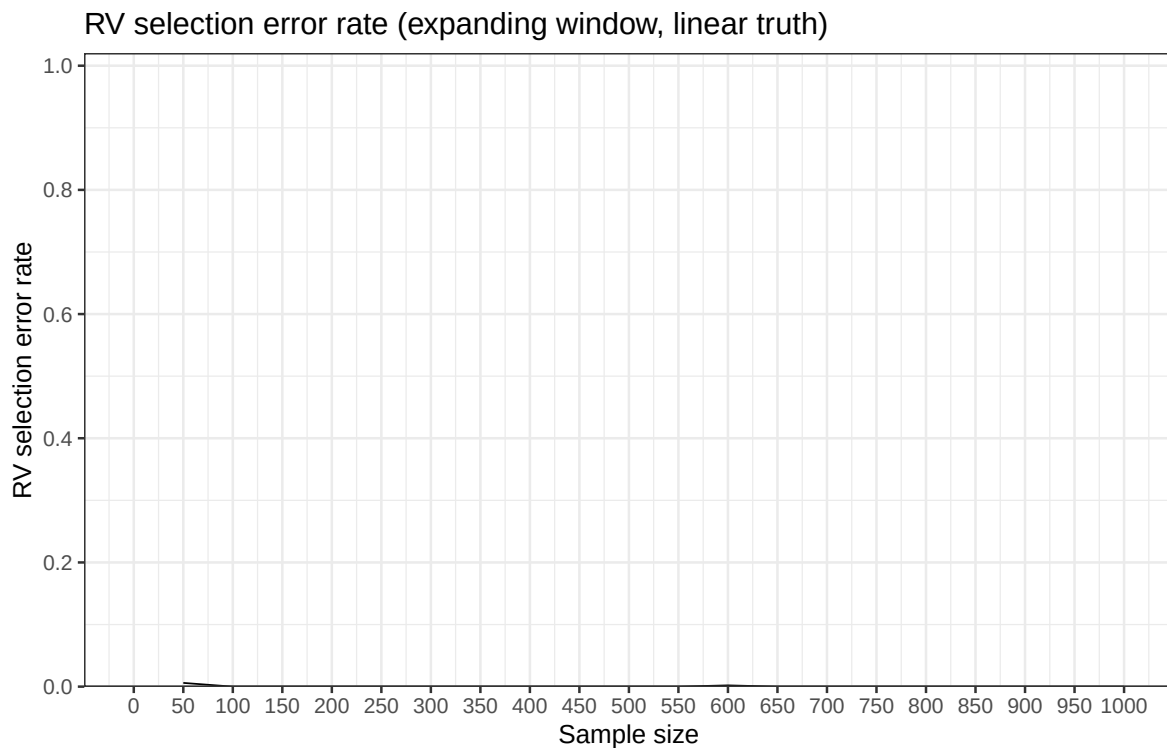


Figure 1: RV selection error rate under linear truth (expanding window).

Polynomial truth

Simulation settings

Table 3: Simulation settings (expanding window, polynomial truth).

setting	value
seed	2026
Monte Carlo replications per n	500
Sample sizes (grid)	50, 100, ..., 1000 (step 50)
truth type	polynomial
beta0	1
beta1	2
beta2	1

setting	value
beta3	1
noise type	gaussian
sigma	2
expanding initial window size	20
horizon	1
validation step	1

Error rate summary

Table 4: Monte Carlo mean selection-error rate (truth is polynomial; error = linear chosen) and approximate MC standard error.

n	error_rate	mc_se
50	0.068	0.01126
100	0.000	0.00000
150	0.000	0.00000
200	0.000	0.00000
250	0.000	0.00000
300	0.000	0.00000
350	0.000	0.00000
400	0.000	0.00000
450	0.000	0.00000
500	0.000	0.00000
550	0.000	0.00000
600	0.000	0.00000
650	0.000	0.00000
700	0.000	0.00000
750	0.000	0.00000
800	0.000	0.00000
850	0.000	0.00000
900	0.000	0.00000
950	0.000	0.00000
1000	0.000	0.00000

Error rate vs sample size

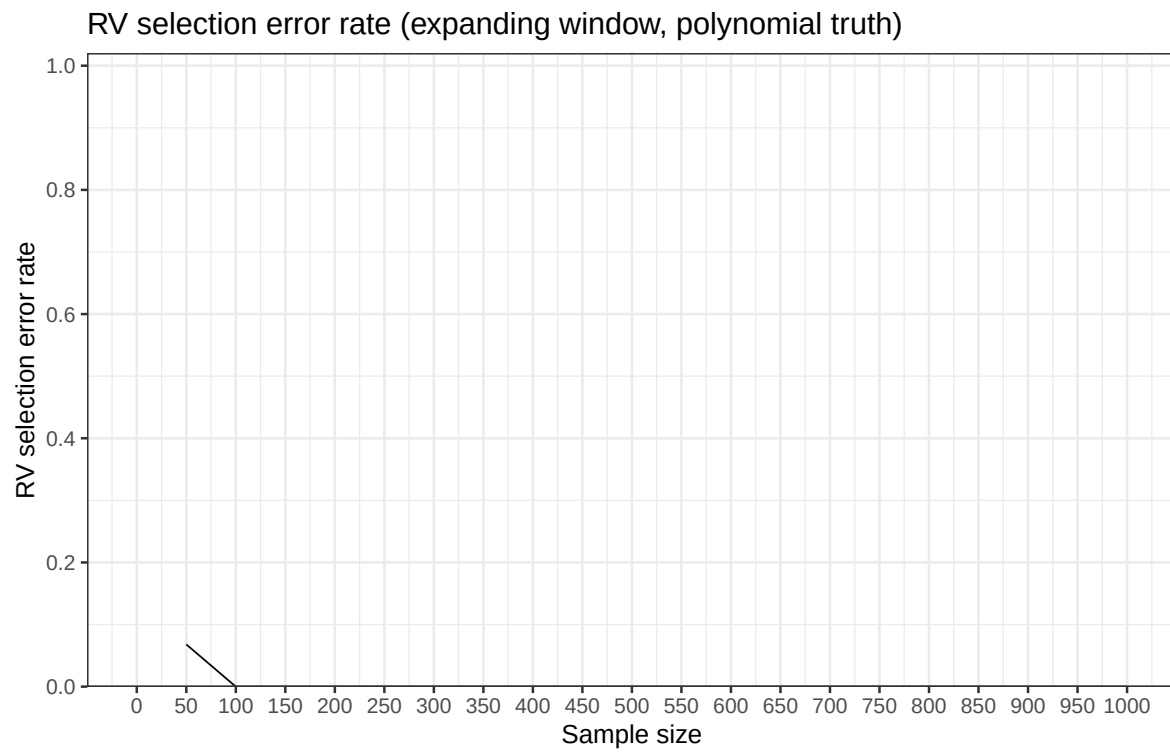


Figure 2: RV selection error rate under polynomial truth (expanding window).